MPG No. 2.2.4 $\bar{1}1'$ [Type II, triclinic] [Q tensor]

* Rank 0 tensor.

$$[Q]$$

$$Q = \mathbb{Q}_0^{(1)}$$

* Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} Q_{xx} & Q_{xy} & Q_{xz} \\ Q_{xy} & Q_{yy} & Q_{yz} \\ Q_{xz} & Q_{yz} & Q_{zz} \end{bmatrix}$$

$$\begin{aligned} Q_{xx} &= \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(1)} + \mathbb{Q}_v^{(1)} \\ Q_{xy} &= \mathbb{Q}_{xy}^{(1)} \\ Q_{xz} &= \mathbb{Q}_{xz}^{(1)} \\ Q_{yy} &= \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(1)} - \mathbb{Q}_v^{(1)} \\ Q_{yz} &= \mathbb{Q}_{yz}^{(1)} \\ Q_{zz} &= \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_u^{(1)} \end{aligned}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & Q_{xy} & Q_{xz} \\ -Q_{xy} & 0 & Q_{yz} \\ -Q_{xz} & -Q_{yz} & 0 \end{bmatrix}$$

$$\begin{aligned} Q_{xy} &= \mathbb{G}_z^{(1)} \\ Q_{xz} &= -\mathbb{G}_y^{(1)} \\ Q_{yz} &= \mathbb{G}_x^{(1)} \end{aligned}$$

* Rank 3 tensor (ss). * Rank 3 tensor (sa). * Rank 3 tensor (a). * Rank 4 tensor (sss).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxyy} & Q_{xxzz} & Q_{xxyz} & Q_{xxzx} & Q_{xxxy} \\ Q_{xxyy} & Q_{yyyy} & Q_{yyzz} & Q_{yyyz} & Q_{yyzx} & Q_{yyxy} \\ Q_{xxzz} & Q_{yyzz} & Q_{zzzz} & Q_{zzyz} & Q_{zzzx} & Q_{zzxy} \\ Q_{xxyz} & Q_{yyyz} & Q_{zzyz} & Q_{yzyz} & Q_{yzzx} & Q_{yzxy} \\ Q_{xxzx} & Q_{yyzx} & Q_{zzzx} & Q_{yzzx} & Q_{zxzx} & Q_{zxxy} \\ Q_{xxxy} & Q_{yyxy} & Q_{zzxy} & Q_{yzxy} & Q_{zxxy} & Q_{xyxy} \end{bmatrix}$$

$$Q_{xxxx} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)} + 2\mathbb{Q}_v^{(1)} + 4\mathbb{Q}_v^{(2)}$$
$$Q_{xxyy} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)}$$
$$Q_{xxzz} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} + \mathbb{Q}_v^{(1)}$$
$$Q_{xxyz} = 2\mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)}$$
$$Q_{xxzx} = -\mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)}$$
$$Q_{xxxy} = \mathbb{Q}_{4z}^{\alpha(1)} - \mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(1)} + 2\mathbb{Q}_{xy}^{(2)}$$
$$Q_{yyyy} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)} - 2\mathbb{Q}_v^{(1)} - 4\mathbb{Q}_v^{(2)}$$
$$Q_{yyzz} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} - \mathbb{Q}_v^{(1)}$$
$$Q_{yyyz} = \mathbb{Q}_{4x}^{\alpha(1)} - \mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + 2\mathbb{Q}_{yz}^{(2)}$$
$$Q_{yyzx} = 2\mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)}$$
$$Q_{yyxy} = -\mathbb{Q}_{4z}^{\alpha(1)} - \mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(1)} + 2\mathbb{Q}_{xy}^{(2)}$$
$$Q_{zzzz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} + 2\mathbb{Q}_{4u}^{(1)} + 2\mathbb{Q}_4^{(1)} + 4\mathbb{Q}_u^{(1)} + 8\mathbb{Q}_u^{(2)}$$
$$Q_{zzyz} = -\mathbb{Q}_{4x}^{\alpha(1)} - \mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + 2\mathbb{Q}_{yz}^{(2)}$$
$$Q_{zzzx} = \mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)}$$
$$Q_{zzxy} = 2\mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(1)}$$
$$Q_{yzyz} = \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} - \mathbb{Q}_v^{(2)}$$
$$Q_{yzzx} = 2\mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(2)}$$
$$Q_{yzxy} = 2\mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(2)}$$
$$Q_{zxzx} = \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} + \mathbb{Q}_v^{(2)}$$
$$Q_{zxxy} = 2\mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(2)}$$
$$Q_{xyxy} = \mathbb{Q}_0^{(2)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & Q_{xxyy} & Q_{xxzz} & Q_{xxyz} & Q_{xxzx} & Q_{xxxy} \\ -Q_{xxyy} & 0 & Q_{yyzz} & Q_{yyyz} & Q_{yyzx} & Q_{yyxy} \\ -Q_{xxzz} & -Q_{yyzz} & 0 & Q_{zzyz} & Q_{zzzx} & Q_{zzxy} \\ -Q_{xxyz} & -Q_{yyyz} & -Q_{zzyz} & 0 & Q_{yzzx} & Q_{yzxy} \\ -Q_{xxzx} & -Q_{yyzx} & -Q_{zzzx} & -Q_{yzzx} & 0 & Q_{zxxy} \\ -Q_{xxxy} & -Q_{yyxy} & -Q_{zzxy} & -Q_{yzxy} & -Q_{zxxy} & 0 \end{bmatrix}$$

$$Q_{xxyy} = 4\mathbb{G}_{xyz}^{(1)} - 2\mathbb{Q}_v^{(3)}$$
$$Q_{xxzz} = -4\mathbb{G}_{xyz}^{(1)} + 3\mathbb{Q}_u^{(3)} - \mathbb{Q}_v^{(3)}$$
$$Q_{xxyz} = -4\mathbb{G}_x^{\beta(1)} + \mathbb{Q}_{yz}^{(3)}$$
$$Q_{xxzx} = -2\mathbb{G}_y^{(1)} + 2\mathbb{G}_y^{\alpha(1)} + 2\mathbb{G}_y^{\beta(1)} + \mathbb{Q}_{xz}^{(3)}$$
$$Q_{xxxy} = 2\mathbb{G}_z^{(1)} - 2\mathbb{G}_z^{\alpha(1)} + 2\mathbb{G}_z^{\beta(1)} + \mathbb{Q}_{xy}^{(3)}$$
$$Q_{yyzz} = 4\mathbb{G}_{xyz}^{(1)} + 3\mathbb{Q}_u^{(3)} + \mathbb{Q}_v^{(3)}$$
$$Q_{yyyz} = 2\mathbb{G}_x^{(1)} - 2\mathbb{G}_x^{\alpha(1)} + 2\mathbb{G}_x^{\beta(1)} + \mathbb{Q}_{yz}^{(3)}$$
$$Q_{yyzx} = -4\mathbb{G}_y^{\beta(1)} + \mathbb{Q}_{xz}^{(3)}$$
$$Q_{yyxy} = -2\mathbb{G}_z^{(1)} + 2\mathbb{G}_z^{\alpha(1)} + 2\mathbb{G}_z^{\beta(1)} + \mathbb{Q}_{xy}^{(3)}$$
$$Q_{zzyz} = -2\mathbb{G}_x^{(1)} + 2\mathbb{G}_x^{\alpha(1)} + 2\mathbb{G}_x^{\beta(1)} + \mathbb{Q}_{yz}^{(3)}$$
$$Q_{zzzx} = 2\mathbb{G}_y^{(1)} - 2\mathbb{G}_y^{\alpha(1)} + 2\mathbb{G}_y^{\beta(1)} + \mathbb{Q}_{xz}^{(3)}$$
$$Q_{zzxy} = -4\mathbb{G}_z^{\beta(1)} + \mathbb{Q}_{xy}^{(3)}$$
$$Q_{yzzx} = -\mathbb{G}_z^{(1)} - 4\mathbb{G}_z^{\alpha(1)}$$
$$Q_{yzxy} = \mathbb{G}_y^{(1)} + 4\mathbb{G}_y^{\alpha(1)}$$
$$Q_{zxxy} = -\mathbb{G}_x^{(1)} - 4\mathbb{G}_x^{\alpha(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} Q_{yzyz} & Q_{yzzx} & Q_{yzxy} \\ Q_{yzzx} & Q_{zxzx} & Q_{zxxy} \\ Q_{yzxy} & Q_{zxxy} & Q_{xyxy} \end{bmatrix}$$

$$Q_{yzyz} = \mathbb{Q}_0^{(3)} - 2\mathbb{Q}_u^{(4)} + 2\mathbb{Q}_v^{(4)}$$
$$Q_{yzzx} = 2\mathbb{Q}_{xy}^{(4)}$$
$$Q_{yzxy} = 2\mathbb{Q}_{xz}^{(4)}$$
$$Q_{zxzx} = \mathbb{Q}_0^{(3)} - 2\mathbb{Q}_u^{(4)} - 2\mathbb{Q}_v^{(4)}$$
$$Q_{zxxy} = 2\mathbb{Q}_{yz}^{(4)}$$
$$Q_{xyxy} = \mathbb{Q}_0^{(3)} + 4\mathbb{Q}_u^{(4)}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & Q_{yzzx} & Q_{yzxy} \\ -Q_{yzzx} & 0 & Q_{zxxy} \\ -Q_{yzxy} & -Q_{zxxy} & 0 \end{bmatrix}$$

$$Q_{yzzx} = -\mathbb{G}_z^{(2)}$$
$$Q_{yzxy} = \mathbb{G}_y^{(2)}$$
$$Q_{zxxy} = -\mathbb{G}_x^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} Q_{xxyz} & Q_{xxzx} & Q_{xxxy} \\ Q_{yyyz} & Q_{yyzx} & Q_{yyxy} \\ Q_{zzyz} & Q_{zzzx} & Q_{zzxy} \\ Q_{yzyz} & Q_{yzzx} & Q_{yzxy} \\ Q_{zxyz} & Q_{zxzx} & Q_{zxxy} \\ Q_{xyyz} & Q_{xyzx} & Q_{xyxy} \end{bmatrix}$$

$$Q_{xxyz} = \mathbb{G}_x^{(3)} + 2\mathbb{G}_x^{(4)} + 2\mathbb{G}_x^{\alpha(2)}$$
$$Q_{xxzx} = \mathbb{G}_y^{(3)} - \mathbb{G}_y^{\alpha(2)} - \mathbb{G}_y^{\beta(2)} + 2\mathbb{Q}_{xz}^{(5)}$$
$$Q_{xxxy} = \mathbb{G}_z^{(3)} - \mathbb{G}_z^{\alpha(2)} + \mathbb{G}_z^{\beta(2)} - 2\mathbb{Q}_{xy}^{(5)}$$
$$Q_{yyyz} = \mathbb{G}_x^{(3)} - \mathbb{G}_x^{\alpha(2)} + \mathbb{G}_x^{\beta(2)} - 2\mathbb{Q}_{yz}^{(5)}$$
$$Q_{yyzx} = \mathbb{G}_y^{(3)} + 2\mathbb{G}_y^{(4)} + 2\mathbb{G}_y^{\alpha(2)}$$
$$Q_{yyxy} = \mathbb{G}_z^{(3)} - \mathbb{G}_z^{\alpha(2)} - \mathbb{G}_z^{\beta(2)} + 2\mathbb{Q}_{xy}^{(5)}$$
$$Q_{zzyz} = \mathbb{G}_x^{(3)} - \mathbb{G}_x^{\alpha(2)} - \mathbb{G}_x^{\beta(2)} + 2\mathbb{Q}_{yz}^{(5)}$$
$$Q_{zzzx} = \mathbb{G}_y^{(3)} - \mathbb{G}_y^{\alpha(2)} + \mathbb{G}_y^{\beta(2)} - 2\mathbb{Q}_{xz}^{(5)}$$
$$Q_{zzxy} = \mathbb{G}_z^{(3)} + 2\mathbb{G}_z^{(4)} + 2\mathbb{G}_z^{\alpha(2)}$$
$$Q_{yzyz} = \mathbb{G}_{xyz}^{(2)} - 3\mathbb{Q}_u^{(5)} - \mathbb{Q}_v^{(5)}$$
$$Q_{yzzx} = \mathbb{G}_z^{(4)} - \mathbb{G}_z^{\alpha(2)} - \mathbb{G}_z^{\beta(2)} - \mathbb{Q}_{xy}^{(5)}$$
$$Q_{yzxy} = \mathbb{G}_y^{(4)} - \mathbb{G}_y^{\alpha(2)} + \mathbb{G}_y^{\beta(2)} + \mathbb{Q}_{xz}^{(5)}$$
$$Q_{zxyz} = \mathbb{G}_z^{(4)} - \mathbb{G}_z^{\alpha(2)} + \mathbb{G}_z^{\beta(2)} + \mathbb{Q}_{xy}^{(5)}$$
$$Q_{zxzx} = \mathbb{G}_{xyz}^{(2)} + 3\mathbb{Q}_u^{(5)} - \mathbb{Q}_v^{(5)}$$
$$Q_{zxxy} = \mathbb{G}_x^{(4)} - \mathbb{G}_x^{\alpha(2)} - \mathbb{G}_x^{\beta(2)} - \mathbb{Q}_{yz}^{(5)}$$
$$Q_{xyyz} = \mathbb{G}_y^{(4)} - \mathbb{G}_y^{\alpha(2)} - \mathbb{G}_y^{\beta(2)} - \mathbb{Q}_{xz}^{(5)}$$
$$Q_{xyzx} = \mathbb{G}_x^{(4)} - \mathbb{G}_x^{\alpha(2)} + \mathbb{G}_x^{\beta(2)} + \mathbb{Q}_{yz}^{(5)}$$
$$Q_{xyxy} = \mathbb{G}_{xyz}^{(2)} + 2\mathbb{Q}_v^{(5)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} Q_{yzxx} & Q_{yzyy} & Q_{yzzz} & Q_{yzyz} & Q_{yzzx} & Q_{yzxy} \\ Q_{zxxx} & Q_{zxyy} & Q_{zxzz} & Q_{zxyz} & Q_{zxzx} & Q_{zxxy} \\ Q_{xyxx} & Q_{xyyy} & Q_{xyzz} & Q_{xyyz} & Q_{xyzx} & Q_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
Q_{yzxx} &= \mathbb{G}_x^{(5)} + 2\mathbb{G}_x^{(6)} + 2\mathbb{G}_x^{\alpha(3)} \\
Q_{yzyy} &= \mathbb{G}_x^{(5)} - \mathbb{G}_x^{\alpha(3)} + \mathbb{G}_x^{\beta(3)} - 2\mathbb{Q}_{yz}^{(6)} \\
Q_{yzzz} &= \mathbb{G}_x^{(5)} - \mathbb{G}_x^{\alpha(3)} - \mathbb{G}_x^{\beta(3)} + 2\mathbb{Q}_{yz}^{(6)} \\
Q_{yzyz} &= \mathbb{G}_{xyz}^{(3)} - 3\mathbb{Q}_u^{(6)} - \mathbb{Q}_v^{(6)} \\
Q_{yzzx} &= \mathbb{G}_z^{(6)} - \mathbb{G}_z^{\alpha(3)} + \mathbb{G}_z^{\beta(3)} + \mathbb{Q}_{xy}^{(6)} \\
Q_{yzxy} &= \mathbb{G}_y^{(6)} - \mathbb{G}_y^{\alpha(3)} - \mathbb{G}_y^{\beta(3)} - \mathbb{Q}_{xz}^{(6)} \\
Q_{zxxx} &= \mathbb{G}_y^{(5)} - \mathbb{G}_y^{\alpha(3)} - \mathbb{G}_y^{\beta(3)} + 2\mathbb{Q}_{xz}^{(6)} \\
Q_{zxyy} &= \mathbb{G}_y^{(5)} + 2\mathbb{G}_y^{(6)} + 2\mathbb{G}_y^{\alpha(3)} \\
Q_{zxzz} &= \mathbb{G}_y^{(5)} - \mathbb{G}_y^{\alpha(3)} + \mathbb{G}_y^{\beta(3)} - 2\mathbb{Q}_{xz}^{(6)} \\
Q_{zxyz} &= \mathbb{G}_z^{(6)} - \mathbb{G}_z^{\alpha(3)} - \mathbb{G}_z^{\beta(3)} - \mathbb{Q}_{xy}^{(6)} \\
Q_{zxzx} &= \mathbb{G}_{xyz}^{(3)} + 3\mathbb{Q}_u^{(6)} - \mathbb{Q}_v^{(6)} \\
Q_{zxxy} &= \mathbb{G}_x^{(6)} - \mathbb{G}_x^{\alpha(3)} + \mathbb{G}_x^{\beta(3)} + \mathbb{Q}_{yz}^{(6)} \\
Q_{xyxx} &= \mathbb{G}_z^{(5)} - \mathbb{G}_z^{\alpha(3)} + \mathbb{G}_z^{\beta(3)} - 2\mathbb{Q}_{xy}^{(6)} \\
Q_{xyyy} &= \mathbb{G}_z^{(5)} - \mathbb{G}_z^{\alpha(3)} - \mathbb{G}_z^{\beta(3)} + 2\mathbb{Q}_{xy}^{(6)} \\
Q_{xyzz} &= \mathbb{G}_z^{(5)} + 2\mathbb{G}_z^{(6)} + 2\mathbb{G}_z^{\alpha(3)} \\
Q_{xyyz} &= \mathbb{G}_y^{(6)} - \mathbb{G}_y^{\alpha(3)} + \mathbb{G}_y^{\beta(3)} + \mathbb{Q}_{xz}^{(6)} \\
Q_{xyzx} &= \mathbb{G}_x^{(6)} - \mathbb{G}_x^{\alpha(3)} - \mathbb{G}_x^{\beta(3)} - \mathbb{Q}_{yz}^{(6)} \\
Q_{xyxy} &= \mathbb{G}_{xyz}^{(3)} + 2\mathbb{Q}_v^{(6)}
\end{aligned}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxyy} & Q_{xxzz} & Q_{xxyz} & Q_{xxzx} & Q_{xxxy} & Q_{xxzy} & Q_{xxxz} & Q_{xxyx} \\ Q_{yyxx} & Q_{yyyy} & Q_{yyzz} & Q_{yyyz} & Q_{yyzx} & Q_{yyxy} & Q_{yyzy} & Q_{yyxz} & Q_{yyyx} \\ Q_{zzxx} & Q_{zzyy} & Q_{zzzz} & Q_{zzyz} & Q_{zzzx} & Q_{zzxy} & Q_{zzzy} & Q_{zzxz} & Q_{zzyx} \\ Q_{yzxx} & Q_{yzyy} & Q_{yzzz} & Q_{yzyz} & Q_{yzzx} & Q_{yzxy} & Q_{yzzy} & Q_{yzxz} & Q_{yzyx} \\ Q_{zxxx} & Q_{zxyy} & Q_{zxzz} & Q_{zxyz} & Q_{zxzx} & Q_{zxxy} & Q_{zxzy} & Q_{zxxz} & Q_{zxyx} \\ Q_{xyxx} & Q_{xyyy} & Q_{xyzz} & Q_{xyyz} & Q_{xyzx} & Q_{xyxy} & Q_{xyzy} & Q_{xyxz} & Q_{xyyx} \end{bmatrix}$$

$$Q_{xxxx} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)} + 2\mathbb{Q}_v^{(1)} + 4\mathbb{Q}_v^{(2)}$$

$$Q_{xxyy} = 4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 2\mathbb{Q}_v^{(3)}$$

$$Q_{xxzz} = -4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} + 3\mathbb{Q}_u^{(3)} + \mathbb{Q}_v^{(1)} - \mathbb{Q}_v^{(3)}$$

$$Q_{xxyz} = \mathbb{G}_x^{(3)} + 2\mathbb{G}_x^{(4)} + 2\mathbb{G}_x^{\alpha(2)} - 4\mathbb{G}_x^{\beta(1)} + 2\mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + \mathbb{Q}_{yz}^{(3)}$$

$$Q_{xxzx} = -2\mathbb{G}_y^{(1)} + \mathbb{G}_y^{(3)} + 2\mathbb{G}_y^{\alpha(1)} - \mathbb{G}_y^{\alpha(2)} + 2\mathbb{G}_y^{\beta(1)} - \mathbb{G}_y^{\beta(2)} - \mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(3)} + 2\mathbb{Q}_{xz}^{(5)}$$

$$Q_{xxxy} = 2\mathbb{G}_z^{(1)} + \mathbb{G}_z^{(3)} - 2\mathbb{G}_z^{\alpha(1)} - \mathbb{G}_z^{\alpha(2)} + 2\mathbb{G}_z^{\beta(1)} + \mathbb{G}_z^{\beta(2)} + \mathbb{Q}_{4z}^{\alpha(1)} - \mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(1)} + 2\mathbb{Q}_{xy}^{(2)} + \mathbb{Q}_{xy}^{(3)} - 2\mathbb{Q}_{xy}^{(5)}$$

$$Q_{xxzy} = -\mathbb{G}_x^{(3)} - 2\mathbb{G}_x^{(4)} - 2\mathbb{G}_x^{\alpha(2)} - 4\mathbb{G}_x^{\beta(1)} + 2\mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + \mathbb{Q}_{yz}^{(3)}$$

$$Q_{xxxz} = -2\mathbb{G}_y^{(1)} - \mathbb{G}_y^{(3)} + 2\mathbb{G}_y^{\alpha(1)} + \mathbb{G}_y^{\alpha(2)} + 2\mathbb{G}_y^{\beta(1)} + \mathbb{G}_y^{\beta(2)} - \mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(3)} - 2\mathbb{Q}_{xz}^{(5)}$$

$$Q_{xxyx} = 2\mathbb{G}_z^{(1)} - \mathbb{G}_z^{(3)} - 2\mathbb{G}_z^{\alpha(1)} + \mathbb{G}_z^{\alpha(2)} + 2\mathbb{G}_z^{\beta(1)} - \mathbb{G}_z^{\beta(2)} + \mathbb{Q}_{4z}^{\alpha(1)} - \mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(1)} + 2\mathbb{Q}_{xy}^{(2)} + \mathbb{Q}_{xy}^{(3)} + 2\mathbb{Q}_{xy}^{(5)}$$

$$Q_{yyxx} = -4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} + 2\mathbb{Q}_v^{(3)}$$

$$Q_{yyyy} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)} - 2\mathbb{Q}_v^{(1)} - 4\mathbb{Q}_v^{(2)}$$

$$Q_{yyzz} = 4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} + 3\mathbb{Q}_u^{(3)} - \mathbb{Q}_v^{(1)} + \mathbb{Q}_v^{(3)}$$

$$Q_{yyyz} = 2\mathbb{G}_x^{(1)} + \mathbb{G}_x^{(3)} - 2\mathbb{G}_x^{\alpha(1)} - \mathbb{G}_x^{\alpha(2)} + 2\mathbb{G}_x^{\beta(1)} + \mathbb{G}_x^{\beta(2)} + \mathbb{Q}_{4x}^{\alpha(1)} - \mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + 2\mathbb{Q}_{yz}^{(2)} + \mathbb{Q}_{yz}^{(3)} - 2\mathbb{Q}_{yz}^{(5)}$$

$$Q_{yyzx} = \mathbb{G}_y^{(3)} + 2\mathbb{G}_y^{(4)} + 2\mathbb{G}_y^{\alpha(2)} - 4\mathbb{G}_y^{\beta(1)} + 2\mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + \mathbb{Q}_{xz}^{(3)}$$

$$Q_{yyxy} = -2\mathbb{G}_z^{(1)} + \mathbb{G}_z^{(3)} + 2\mathbb{G}_z^{\alpha(1)} - \mathbb{G}_z^{\alpha(2)} + 2\mathbb{G}_z^{\beta(1)} - \mathbb{G}_z^{\beta(2)} - \mathbb{Q}_{4z}^{\alpha(1)} - \mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(1)} + 2\mathbb{Q}_{xy}^{(2)} + \mathbb{Q}_{xy}^{(3)} + 2\mathbb{Q}_{xy}^{(5)}$$

$$Q_{yyzy} = 2\mathbb{G}_x^{(1)} - \mathbb{G}_x^{(3)} - 2\mathbb{G}_x^{\alpha(1)} + \mathbb{G}_x^{\alpha(2)} + 2\mathbb{G}_x^{\beta(1)} - \mathbb{G}_x^{\beta(2)} + \mathbb{Q}_{4x}^{\alpha(1)} - \mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + 2\mathbb{Q}_{yz}^{(2)} + \mathbb{Q}_{yz}^{(3)} + 2\mathbb{Q}_{yz}^{(5)}$$

$$Q_{yyxz} = -\mathbb{G}_y^{(3)} - 2\mathbb{G}_y^{(4)} - 2\mathbb{G}_y^{\alpha(2)} - 4\mathbb{G}_y^{\beta(1)} + 2\mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + \mathbb{Q}_{xz}^{(3)}$$

$$Q_{yyyx} = -2\mathbb{G}_z^{(1)} - \mathbb{G}_z^{(3)} + 2\mathbb{G}_z^{\alpha(1)} + \mathbb{G}_z^{\alpha(2)} + 2\mathbb{G}_z^{\beta(1)} + \mathbb{G}_z^{\beta(2)} - \mathbb{Q}_{4z}^{\alpha(1)} - \mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(1)} + 2\mathbb{Q}_{xy}^{(2)} + \mathbb{Q}_{xy}^{(3)} - 2\mathbb{Q}_{xy}^{(5)}$$

$$Q_{zzxx} = 4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} - 3\mathbb{Q}_u^{(3)} + \mathbb{Q}_v^{(1)} + \mathbb{Q}_v^{(3)}$$

$$Q_{zzyy} = -4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} - 3\mathbb{Q}_u^{(3)} - \mathbb{Q}_v^{(1)} - \mathbb{Q}_v^{(3)}$$

$$Q_{zzzz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} + 2\mathbb{Q}_{4u}^{(1)} + 2\mathbb{Q}_4^{(1)} + 4\mathbb{Q}_u^{(1)} + 8\mathbb{Q}_u^{(2)}$$

$$Q_{zzyz} = -2\mathbb{G}_x^{(1)} + \mathbb{G}_x^{(3)} + 2\mathbb{G}_x^{\alpha(1)} - \mathbb{G}_x^{\alpha(2)} + 2\mathbb{G}_x^{\beta(1)} - \mathbb{G}_x^{\beta(2)} - \mathbb{Q}_{4x}^{\alpha(1)} - \mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + 2\mathbb{Q}_{yz}^{(2)} + \mathbb{Q}_{yz}^{(3)} + 2\mathbb{Q}_{yz}^{(5)}$$

$$Q_{zzzx} = 2\mathbb{G}_y^{(1)} + \mathbb{G}_y^{(3)} - 2\mathbb{G}_y^{\alpha(1)} - \mathbb{G}_y^{\alpha(2)} + 2\mathbb{G}_y^{\beta(1)} + \mathbb{G}_y^{\beta(2)} + \mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(3)} - 2\mathbb{Q}_{xz}^{(5)}$$

$$Q_{zzxy} = \mathbb{G}_z^{(3)} + 2\mathbb{G}_z^{(4)} + 2\mathbb{G}_z^{\alpha(2)} - 4\mathbb{G}_z^{\beta(1)} + 2\mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(1)} + \mathbb{Q}_{xy}^{(3)}$$

$$Q_{zzzy} = -2\mathbb{G}_x^{(1)} - \mathbb{G}_x^{(3)} + 2\mathbb{G}_x^{\alpha(1)} + \mathbb{G}_x^{\alpha(2)} + 2\mathbb{G}_x^{\beta(1)} + \mathbb{G}_x^{\beta(2)} - \mathbb{Q}_{4x}^{\alpha(1)} - \mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + 2\mathbb{Q}_{yz}^{(2)} + \mathbb{Q}_{yz}^{(3)} - 2\mathbb{Q}_{yz}^{(5)}$$

$$Q_{zzxz} = 2\mathbb{G}_y^{(1)} - \mathbb{G}_y^{(3)} - 2\mathbb{G}_y^{\alpha(1)} + \mathbb{G}_y^{\alpha(2)} + 2\mathbb{G}_y^{\beta(1)} - \mathbb{G}_y^{\beta(2)} + \mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(3)} + 2\mathbb{Q}_{xz}^{(5)}$$

$$Q_{zzyx} = -\mathbb{G}_z^{(3)} - 2\mathbb{G}_z^{(4)} - 2\mathbb{G}_z^{\alpha(2)} - 4\mathbb{G}_z^{\beta(1)} + 2\mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(1)} + \mathbb{Q}_{xy}^{(3)}$$

$$Q_{yzxx} = 4\mathbb{G}_x^{\beta(1)} + 2\mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} - \mathbb{Q}_{yz}^{(3)}$$

$$Q_{yzyy} = -2\mathbb{G}_x^{(1)} + 2\mathbb{G}_x^{\alpha(1)} - 2\mathbb{G}_x^{\beta(1)} + \mathbb{Q}_{4x}^{\alpha(1)} - \mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + 2\mathbb{Q}_{yz}^{(2)} - \mathbb{Q}_{yz}^{(3)}$$

$$Q_{yzzz} = 2\mathbb{G}_x^{(1)} - 2\mathbb{G}_x^{\alpha(1)} - 2\mathbb{G}_x^{\beta(1)} - \mathbb{Q}_{4x}^{\alpha(1)} - \mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + 2\mathbb{Q}_{yz}^{(2)} - \mathbb{Q}_{yz}^{(3)}$$

$$Q_{yzyz} = \mathbb{G}_{xyz}^{(2)} + \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} - 3\mathbb{Q}_u^{(5)} - \mathbb{Q}_v^{(2)} - \mathbb{Q}_v^{(5)}$$

$$Q_{yzzx} = -\mathbb{G}_z^{(1)} + \mathbb{G}_z^{(4)} - 4\mathbb{G}_z^{\alpha(1)} - \mathbb{G}_z^{\alpha(2)} - \mathbb{G}_z^{\beta(2)} + 2\mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(2)} - \mathbb{Q}_{xy}^{(5)}$$

$$Q_{yzxy} = \mathbb{G}_y^{(1)} + \mathbb{G}_y^{(4)} + 4\mathbb{G}_y^{\alpha(1)} - \mathbb{G}_y^{\alpha(2)} + \mathbb{G}_y^{\beta(2)} + 2\mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(5)}$$

$$Q_{yzzy} = -\mathbb{G}_{xyz}^{(2)} + \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} + 3\mathbb{Q}_u^{(5)} - \mathbb{Q}_v^{(2)} + \mathbb{Q}_v^{(5)}$$

$$Q_{yzxz} = -\mathbb{G}_z^{(1)} - \mathbb{G}_z^{(4)} - 4\mathbb{G}_z^{\alpha(1)} + \mathbb{G}_z^{\alpha(2)} + \mathbb{G}_z^{\beta(2)} + 2\mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(2)} + \mathbb{Q}_{xy}^{(5)}$$

$$Q_{yzyx} = \mathbb{G}_y^{(1)} - \mathbb{G}_y^{(4)} + 4\mathbb{G}_y^{\alpha(1)} + \mathbb{G}_y^{\alpha(2)} - \mathbb{G}_y^{\beta(2)} + 2\mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(2)} - \mathbb{Q}_{xz}^{(5)}$$

$$Q_{zxxx} = 2\mathbb{G}_y^{(1)} - 2\mathbb{G}_y^{\alpha(1)} - 2\mathbb{G}_y^{\beta(1)} - \mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} - \mathbb{Q}_{xz}^{(3)}$$

$$Q_{zxyy} = 4\mathbb{G}_y^{\beta(1)} + 2\mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} - \mathbb{Q}_{xz}^{(3)}$$

$$Q_{zxzz} = -2\mathbb{G}_y^{(1)} + 2\mathbb{G}_y^{\alpha(1)} - 2\mathbb{G}_y^{\beta(1)} + \mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} - \mathbb{Q}_{xz}^{(3)}$$

$$Q_{zxyz} = \mathbb{G}_z^{(1)} + \mathbb{G}_z^{(4)} + 4\mathbb{G}_z^{\alpha(1)} - \mathbb{G}_z^{\alpha(2)} + \mathbb{G}_z^{\beta(2)} + 2\mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(2)} + \mathbb{Q}_{xy}^{(5)}$$

$$Q_{zxzx} = \mathbb{G}_{xyz}^{(2)} + \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} + 3\mathbb{Q}_u^{(5)} + \mathbb{Q}_v^{(2)} - \mathbb{Q}_v^{(5)}$$

$$Q_{zxxy} = -\mathbb{G}_x^{(1)} + \mathbb{G}_x^{(4)} - 4\mathbb{G}_x^{\alpha(1)} - \mathbb{G}_x^{\alpha(2)} - \mathbb{G}_x^{\beta(2)} + 2\mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(2)} - \mathbb{Q}_{yz}^{(5)}$$

$$Q_{zxzy} = \mathbb{G}_z^{(1)} - \mathbb{G}_z^{(4)} + 4\mathbb{G}_z^{\alpha(1)} + \mathbb{G}_z^{\alpha(2)} - \mathbb{G}_z^{\beta(2)} + 2\mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(2)} - \mathbb{Q}_{xy}^{(5)}$$

$$Q_{zxxz} = -\mathbb{G}_{xyz}^{(2)} + \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} - 3\mathbb{Q}_u^{(5)} + \mathbb{Q}_v^{(2)} + \mathbb{Q}_v^{(5)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} Q_{yzxx} & Q_{yzyy} & Q_{yzzz} & Q_{yzyz} & Q_{yzzx} & Q_{yzxy} & Q_{yzzy} & Q_{yzxz} & Q_{yzyx} \\ Q_{zxxx} & Q_{zxyy} & Q_{zxzz} & Q_{zxyz} & Q_{zxzx} & Q_{zxxy} & Q_{zxzy} & Q_{zxxz} & Q_{zxyx} \\ Q_{xyxx} & Q_{xyyy} & Q_{xyzz} & Q_{xyyz} & Q_{xyzx} & Q_{xyxy} & Q_{xyzy} & Q_{xyxz} & Q_{xyyx} \end{bmatrix}$$

$$\begin{aligned}
Q_{yzxx} &= \mathbb{G}_x^{(5)} + 2\mathbb{G}_x^{(6)} + 2\mathbb{G}_x^{\alpha(3)} \\
Q_{yzyy} &= \mathbb{G}_x^{(5)} - \mathbb{G}_x^{\alpha(3)} + \mathbb{G}_x^{\beta(3)} - 2\mathbb{Q}_{yz}^{(6)} \\
Q_{yzzz} &= \mathbb{G}_x^{(5)} - \mathbb{G}_x^{\alpha(3)} - \mathbb{G}_x^{\beta(3)} + 2\mathbb{Q}_{yz}^{(6)} \\
Q_{yzyz} &= \mathbb{G}_{xyz}^{(3)} + \mathbb{Q}_0^{(3)} - 2\mathbb{Q}_u^{(4)} - 3\mathbb{Q}_u^{(6)} + 2\mathbb{Q}_v^{(4)} - \mathbb{Q}_v^{(6)} \\
Q_{yzzx} &= -\mathbb{G}_z^{(2)} + \mathbb{G}_z^{(6)} - \mathbb{G}_z^{\alpha(3)} + \mathbb{G}_z^{\beta(3)} + 2\mathbb{Q}_{xy}^{(4)} + \mathbb{Q}_{xy}^{(6)} \\
Q_{yzxy} &= \mathbb{G}_y^{(2)} + \mathbb{G}_y^{(6)} - \mathbb{G}_y^{\alpha(3)} - \mathbb{G}_y^{\beta(3)} + 2\mathbb{Q}_{xz}^{(4)} - \mathbb{Q}_{xz}^{(6)} \\
Q_{yzzy} &= \mathbb{G}_{xyz}^{(3)} - \mathbb{Q}_0^{(3)} + 2\mathbb{Q}_u^{(4)} - 3\mathbb{Q}_u^{(6)} - 2\mathbb{Q}_v^{(4)} - \mathbb{Q}_v^{(6)} \\
Q_{yzxz} &= \mathbb{G}_z^{(2)} + \mathbb{G}_z^{(6)} - \mathbb{G}_z^{\alpha(3)} + \mathbb{G}_z^{\beta(3)} - 2\mathbb{Q}_{xy}^{(4)} + \mathbb{Q}_{xy}^{(6)} \\
Q_{yzyx} &= -\mathbb{G}_y^{(2)} + \mathbb{G}_y^{(6)} - \mathbb{G}_y^{\alpha(3)} - \mathbb{G}_y^{\beta(3)} - 2\mathbb{Q}_{xz}^{(4)} - \mathbb{Q}_{xz}^{(6)} \\
Q_{zxxx} &= \mathbb{G}_y^{(5)} - \mathbb{G}_y^{\alpha(3)} - \mathbb{G}_y^{\beta(3)} + 2\mathbb{Q}_{xz}^{(6)} \\
Q_{zxyy} &= \mathbb{G}_y^{(5)} + 2\mathbb{G}_y^{(6)} + 2\mathbb{G}_y^{\alpha(3)} \\
Q_{zxzz} &= \mathbb{G}_y^{(5)} - \mathbb{G}_y^{\alpha(3)} + \mathbb{G}_y^{\beta(3)} - 2\mathbb{Q}_{xz}^{(6)} \\
Q_{zxyz} &= \mathbb{G}_z^{(2)} + \mathbb{G}_z^{(6)} - \mathbb{G}_z^{\alpha(3)} - \mathbb{G}_z^{\beta(3)} + 2\mathbb{Q}_{xy}^{(4)} - \mathbb{Q}_{xy}^{(6)} \\
Q_{zxzx} &= \mathbb{G}_{xyz}^{(3)} + \mathbb{Q}_0^{(3)} - 2\mathbb{Q}_u^{(4)} + 3\mathbb{Q}_u^{(6)} - 2\mathbb{Q}_v^{(4)} - \mathbb{Q}_v^{(6)} \\
Q_{zxxy} &= -\mathbb{G}_x^{(2)} + \mathbb{G}_x^{(6)} - \mathbb{G}_x^{\alpha(3)} + \mathbb{G}_x^{\beta(3)} + 2\mathbb{Q}_{yz}^{(4)} + \mathbb{Q}_{yz}^{(6)} \\
Q_{zxzy} &= -\mathbb{G}_z^{(2)} + \mathbb{G}_z^{(6)} - \mathbb{G}_z^{\alpha(3)} - \mathbb{G}_z^{\beta(3)} - 2\mathbb{Q}_{xy}^{(4)} - \mathbb{Q}_{xy}^{(6)} \\
Q_{zxxz} &= \mathbb{G}_{xyz}^{(3)} - \mathbb{Q}_0^{(3)} + 2\mathbb{Q}_u^{(4)} + 3\mathbb{Q}_u^{(6)} + 2\mathbb{Q}_v^{(4)} - \mathbb{Q}_v^{(6)} \\
Q_{zxyx} &= \mathbb{G}_x^{(2)} + \mathbb{G}_x^{(6)} - \mathbb{G}_x^{\alpha(3)} + \mathbb{G}_x^{\beta(3)} - 2\mathbb{Q}_{yz}^{(4)} + \mathbb{Q}_{yz}^{(6)} \\
Q_{xyxx} &= \mathbb{G}_z^{(5)} - \mathbb{G}_z^{\alpha(3)} + \mathbb{G}_z^{\beta(3)} - 2\mathbb{Q}_{xy}^{(6)} \\
Q_{xyyy} &= \mathbb{G}_z^{(5)} - \mathbb{G}_z^{\alpha(3)} - \mathbb{G}_z^{\beta(3)} + 2\mathbb{Q}_{xy}^{(6)} \\
Q_{xyzz} &= \mathbb{G}_z^{(5)} + 2\mathbb{G}_z^{(6)} + 2\mathbb{G}_z^{\alpha(3)} \\
Q_{xyyz} &= -\mathbb{G}_y^{(2)} + \mathbb{G}_y^{(6)} - \mathbb{G}_y^{\alpha(3)} + \mathbb{G}_y^{\beta(3)} + 2\mathbb{Q}_{xz}^{(4)} + \mathbb{Q}_{xz}^{(6)} \\
Q_{xyzx} &= \mathbb{G}_x^{(2)} + \mathbb{G}_x^{(6)} - \mathbb{G}_x^{\alpha(3)} - \mathbb{G}_x^{\beta(3)} + 2\mathbb{Q}_{yz}^{(4)} - \mathbb{Q}_{yz}^{(6)} \\
Q_{xyxy} &= \mathbb{G}_{xyz}^{(3)} + \mathbb{Q}_0^{(3)} + 4\mathbb{Q}_u^{(4)} + 2\mathbb{Q}_v^{(6)} \\
Q_{xyzy} &= \mathbb{G}_y^{(2)} + \mathbb{G}_y^{(6)} - \mathbb{G}_y^{\alpha(3)} + \mathbb{G}_y^{\beta(3)} - 2\mathbb{Q}_{xz}^{(4)} + \mathbb{Q}_{xz}^{(6)} \\
Q_{xyxz} &= -\mathbb{G}_x^{(2)} + \mathbb{G}_x^{(6)} - \mathbb{G}_x^{\alpha(3)} - \mathbb{G}_x^{\beta(3)} - 2\mathbb{Q}_{yz}^{(4)} - \mathbb{Q}_{yz}^{(6)} \\
Q_{xyyx} &= \mathbb{G}_{xyz}^{(3)} - \mathbb{Q}_0^{(3)} - 4\mathbb{Q}_u^{(4)} + 2\mathbb{Q}_v^{(6)}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxxy} & Q_{xxxz} \\ Q_{yyyx} & Q_{yyyy} & Q_{yyyz} \\ Q_{zzzx} & Q_{zzzy} & Q_{zzzz} \\ Q_{yyzx} & Q_{yyzy} & Q_{yyzz} \\ Q_{zzxx} & Q_{zzxy} & Q_{zzxz} \\ Q_{xxyx} & Q_{xxyy} & Q_{xxyz} \\ Q_{yzzx} & Q_{yzzy} & Q_{yzzz} \\ Q_{zxxx} & Q_{zxxy} & Q_{zxxz} \\ Q_{xxyy} + Q_{yyzz} - Q_{yzzy} - Q_{zxxz} + Q_{zzxx} & Q_{xyyy} & Q_{xyyz} \\ -\frac{Q_{yyyz}}{2} + \frac{Q_{yyzy}}{2} + Q_{zxxy} & Q_{xyyz} + \frac{Q_{zzzz}}{2} - \frac{Q_{zzzx}}{2} & \frac{Q_{xxxy}}{2} - \frac{Q_{xxyx}}{2} + Q_{xyyy} - Q_{yyyx} + Q_{zzxy} \end{bmatrix}$$

$$Q_{xxxx} = 3\mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)} + 2\mathbb{Q}_v^{(1)} + 4\mathbb{Q}_v^{(2)}$$
$$Q_{xxxy} = \mathbb{Q}_{4z}^{\alpha(1)} - \mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(1)} + 2\mathbb{Q}_{xy}^{(2)} + \mathbb{Q}_{xy}^{(3)} - 2\mathbb{Q}_{xy}^{(5)}$$
$$Q_{xxxz} = -\mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(3)} - 2\mathbb{Q}_{xz}^{(5)}$$
$$Q_{yyyx} = -\mathbb{Q}_{4z}^{\alpha(1)} - \mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(1)} + 2\mathbb{Q}_{xy}^{(2)} + \mathbb{Q}_{xy}^{(3)} - 2\mathbb{Q}_{xy}^{(5)}$$
$$Q_{yyyy} = 3\mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)} - 2\mathbb{Q}_v^{(1)} - 4\mathbb{Q}_v^{(2)}$$
$$Q_{yyyz} = \mathbb{Q}_{4x}^{\alpha(1)} - \mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + 2\mathbb{Q}_{yz}^{(2)} + \mathbb{Q}_{yz}^{(3)} - 2\mathbb{Q}_{yz}^{(5)}$$
$$Q_{zzzx} = \mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(3)} - 2\mathbb{Q}_{xz}^{(5)}$$
$$Q_{zzzy} = -\mathbb{Q}_{4x}^{\alpha(1)} - \mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + 2\mathbb{Q}_{yz}^{(2)} + \mathbb{Q}_{yz}^{(3)} - 2\mathbb{Q}_{yz}^{(5)}$$
$$Q_{zzzz} = 3\mathbb{Q}_0^{(1)} + 2\mathbb{Q}_{4u}^{(1)} + 2\mathbb{Q}_4^{(1)} + 4\mathbb{Q}_u^{(1)} + 8\mathbb{Q}_u^{(2)}$$
$$Q_{yyzx} = 2\mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + \mathbb{Q}_{xz}^{(3)}$$
$$Q_{yyzy} = \mathbb{Q}_{4x}^{\alpha(1)} - \mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + 2\mathbb{Q}_{yz}^{(2)} + \mathbb{Q}_{yz}^{(3)} + 2\mathbb{Q}_{yz}^{(5)}$$
$$Q_{yyzz} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} + 3\mathbb{Q}_u^{(3)} - \mathbb{Q}_v^{(1)} + \mathbb{Q}_v^{(3)}$$
$$Q_{zzxx} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} - 3\mathbb{Q}_u^{(3)} + \mathbb{Q}_v^{(1)} + \mathbb{Q}_v^{(3)}$$
$$Q_{zzxy} = 2\mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(1)} + \mathbb{Q}_{xy}^{(3)}$$
$$Q_{zzxz} = \mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(3)} + 2\mathbb{Q}_{xz}^{(5)}$$
$$Q_{xxyx} = \mathbb{Q}_{4z}^{\alpha(1)} - \mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(1)} + 2\mathbb{Q}_{xy}^{(2)} + \mathbb{Q}_{xy}^{(3)} + 2\mathbb{Q}_{xy}^{(5)}$$
$$Q_{xxyy} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 2\mathbb{Q}_v^{(3)}$$
$$Q_{xxyz} = 2\mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + \mathbb{Q}_{yz}^{(3)}$$
$$Q_{yzzx} = 2\mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(2)} - \mathbb{Q}_{xy}^{(5)}$$
$$Q_{yzzy} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} + 3\mathbb{Q}_u^{(5)} - \mathbb{Q}_v^{(2)} + \mathbb{Q}_v^{(5)}$$
$$Q_{yzzz} = -\mathbb{Q}_{4x}^{\alpha(1)} - \mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(1)} + 2\mathbb{Q}_{yz}^{(2)} - \mathbb{Q}_{yz}^{(3)}$$
$$Q_{zxxx} = -\mathbb{Q}_{4y}^{\alpha(1)} - \mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} - \mathbb{Q}_{xz}^{(3)}$$
$$Q_{zxxy} = 2\mathbb{Q}_{4x}^{\beta(1)} + \mathbb{Q}_{yz}^{(2)} - \mathbb{Q}_{yz}^{(5)}$$
$$Q_{zxxz} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} - 3\mathbb{Q}_u^{(5)} + \mathbb{Q}_v^{(2)} + \mathbb{Q}_v^{(5)}$$
$$Q_{xyyy} = -\mathbb{Q}_{4z}^{\alpha(1)} - \mathbb{Q}_{4z}^{\beta(1)} + \mathbb{Q}_{xy}^{(1)} + 2\mathbb{Q}_{xy}^{(2)} - \mathbb{Q}_{xy}^{(3)}$$
$$Q_{xyyz} = 2\mathbb{Q}_{4y}^{\beta(1)} + \mathbb{Q}_{xz}^{(2)} - \mathbb{Q}_{xz}^{(5)}$$